Problem Set 3

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: ILA 2.6, 2.7, 3.1, 3.3. Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from ILA, 5th edition (Strang, 2016).

1. ILA, Problem Set 2.4 (rules for matrix operations): problems 1, 6, 12.
2. ILA, Problem Set 2.6 (elimination = factorization $A = LU$): problems 1, 5, 13.
3. ILA, Problem Set 2.7 (transposes and permutations): problems 1, 5.
4. ILA, Problem Set 3.1 (spaces of vectors): problems 1, 4, 9.
5. ILA, Problem Set 3.3 (rank and the complete solution to $Ax = b$): problems 1, 5.
6. **[code]** Implement the LU factorization of a square matrix and use it to solve a linear system $Ax = b$. Verify that $LU = A$ (or $PA = LU$ if you pivot).
7. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).